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PROCESS DESIGN GUIDELINES

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1 APPLICATION

This specification will be applied to all process units and relevant system on the Power Plant project of Karabatan Utility Solutions (KUS) located on Karabatan area, Atyrau region, Kazakhstan.

Additions or revisions to the specification may be proposed by Licensors and Subcontractors by raising a Technical Query.

The objective of this document is to ensure consistency of approach to design of equipment and bulk items across the Project.

2 GENERAL PRINCIPLE

This philosophy established in this document shall apply to the sizing of equipment and bulk items specified for the Project.

3 REFERENCED DOCUMENTS

Document Number	Title
01000-EM-SPC-0003	General Site Information: Meteorological, Geological and Utilities Data

4 EQUIPMENT SIZING CRITERIA

4.1 Vessels

4.1.1 Drum Surge Capacity and Hold-Up Volume

For drums the surge capacity (hold-up) is defined as the volume between high and low liquid levels.

If a uniform discharge rate is important, provide the general hold-up times based on design flowrates below:

Service		Hold-up time (Minutes)
Feed to Column or Furnace (feed surge) Drum diameter, metres	below 1.2	20
	from 1.2 to 1.8, inclusive	15
	above 1.8	10

Reflux to Column	5
Product to Storage	2
Flow to Heat Exchanger	2
Flow to Sewer or Drain	1

In case hold-up must be provided for both product and reflux, the larger volume is used, not the sum of the two volumes.

When the discharge rate is unimportant, a nominal hold-up time of approximately two (2) minutes shall be provided.

The normal operating level should be taken as the midpoint between the high and low levels. Level control should span between the high and low levels. Where high and low level trips are required, these should be located at a reasonable elevation above and below the high and low levels respectively, to allow operator intervention before a trip occurs.

Low liquid level shall be at least 200 mm above the bottom (for horizontal vessels) or bottom tangent line (for vertical vessels).

For water settling the above applies but this may need to be increased. Where there may be solids in the drum, which are not to be drawn off, the liquid outlet may be raised and the low liquid level shall be increased accordingly. Provide 5 minutes holdup at the total hydrocarbon rate below the low liquid level for the "settling out of water. In Case no pot is employed, holdup for the water itself also be provided in the bottom of the drum.

For liquid/ liquid separation drum, drum size is a function of settling rate and flow rate. The volume must be such as to provide sufficient residence time for a drop move from the top or bottom of the drum to the interface of the two fluids and coalesce.

For steam drum, holdup time shall be defined by vendor.

For horizontal vessels the HLL is either 300 mm or 20% of the drum diameter from the top, whichever is the greater. Note: if a crinkled wire mesh pad is present then HLL is 300mm from the bottom of the pad.

For vertical vessels, if vapour flow is present the highest liquid level shall be at least 300mm below the bottom of the inlet arrangement. If little or no vapour is present, the highest liquid level shall be at least 300 mm or 15% of the drum diameter below the top tangent line, whichever is the greater.

The minimum time between a HHLL alarm and trip and between a LLLL alarm and trip shall be no less than 2 minutes.

4.1.2 Column Bottoms Surge Capacity and Hold-Up Volume

The liquid residence time (from the low to high levels) for the design of the bottom section of a column is as follows:

- 1) Bottoms as feed to a subsequent column on level control is five (5) minutes. In general, level control will frequently prove satisfactory to the second of a series of columns.
- 2) Bottoms as feed to a subsequent tower on flow control is ten (10) to twenty (20) minutes, when the column is acting as a feed surge drum to another unit. This surge capacity may be obtained by swaging to a larger diameter for the hold-up section of the column, in some cases.
- 3) Bottoms to a heat exchanger and/or tankage is two (2) minutes. This may be reduced in order to prevent fouling in the column.

4.1.3 Knockout Drum Surge Capacity and Hold-Up Volume

For normal accumulation the following liquid hold up applies:

- 1) At low normal accumulation rate

Liquid draw-off is usually manually controlled. Enough volume should be provided to ensure the frequency of emptying is less than once per shift (i.e. eight (8) hours) or preferably twenty four (24) hours. Generally a nominal height above the lower tangent line (say 200mm) will be adequate.

- 2) At higher normal accumulation rate

Liquid draw-off is usually under level control. The distance between high and low level is usually made to suit a standard controller range, (corresponding to controller connections and generally providing hold up time far in excess of the normal requirement of approximately two (2) minutes).

For a spill the following liquid hold up applies:

- 3) Frequently capacity is required for "spill from preceding unit". Provide a volume equal to the entire production of the unit for ten (10) minutes between the 'alarm' level and a point 300 mm below low point of the feed inlet internals.

Sometimes spill requirements govern the drum design. That is, an L/D ratio of approximately 3:1 results in a large drum diameter relative to the vapour load. Use of a crinkle wire mesh screen would then be unnecessary.

4.1.4 Minimum Auxiliary Nozzle Size

The diameter of the recovery unit should be chosen depending on the speed (flow) and the recommended pumping rate of liquid flow in pipes. The diameter of the drain pipe is made in accordance with the hydraulic calculations based on the conditions of the regular time of drainage, but not less than 25mm.

The following list is a recommendation to the minimum auxiliary nozzle sizes to be used for process design sizing of nozzles

Vessel Volume, m3	Vent	Drain	Pump out	Steam out	Blowdown
1.5	25 (1")	25 (1")	25 (1")	25 (1")	50 (2")
1.5-5.6	25 (1")	40(1 1/2")	40(1 1/2")	25 (1")	80 (3")
5.6-17	50 (2")	50 (2")	50 (2")	25 (1")	80 (3")
17-70	50 (2")	80 (3")	80 (3")	50 (2")	100 (4")
70 and over	50 (2")	80 (3")	80 (3")	80 (3")	100 (4")

4.2 Columns

4.2.1 Minimum Tray Spacing

Tower ID mm	Max No. of Tray Passes	Min Tray Spacing mm
750 to 1800	1	500
1800 to 2700	2	500
2700 to 3300	2	600
3300 to 4800	4	600
4800 to 6000	4	600
>6000	4	750

For draw off trays, the spacing is set by the draw off tray design, including hold up.

4.2.2 L/D Ratio

Towers with L/D greater than 25 shall be avoided, if possible, due to support problems.

4.2.3 Allowable Pressure Drop Guidelines

Tray type	In Pressure service (bar per tray)
Sieve	0.007-0.014
Valve	0.007-0.014

4.2.4 Foaming Factor Guidelines

Systems	Foaming factor
Non foaming	1
Low foaming tendencies	0.9
Moderate foaming tendencies	0.75
High foaming tendencies	0.6

At the stage of design for construction, the foaming factors shall be confirmed with the manufacturer of trays or packing as required.

4.2.5 Flooding Factors Guidelines

Valve tray columns will be specified with the following maximum flooding factors:

Systems	Flooding factor
for vacuum towers	77
for other services	82
for column diameters under 900 mm	70

4.2.6 Column Internals

Valve trays shall be the preferred choice. High capacity trays should generally be avoided unless justified on economic grounds.

Trays shall be numbered from the top to the bottom of a column.

All column trays in fouling services shall be provided with centrally positioned manway to enable a high pressure water cleaning device to be lowered down the column. If downcomers are fitted adequate provision must be made to prevent foulant blocking the downcomers. Each column shall be provided with at least one steam-out nozzle, to be located at the bottom of the column.

Vortex breakers shall be installed in the liquid outlets of all liquid-containing columns and vessels, in the case a pump is installed on the outlet liquid circuit. This includes column bottoms, column draw-off trays, accumulators, knock-out drums, etc.

4.3 Heat Exchangers

4.3.1 Fouling Factors

Site specific fouling factors shall be set as shown below. However, where Licensor has requirements that are more stringent than these, then Licensor values shall be applied.

- 1) Closed Circuit Cooling water: $0.00013 \text{ m}^2 \text{ }^\circ\text{C} / \text{W}$
- 2) Cooling water in open recirculation loop: $0.00034 \text{ m}^2 \text{ }^\circ\text{C} / \text{W}$
- 3) For steam services $0.00086 \text{ m}^2 \text{ }^\circ\text{C} / \text{W}$.
- 4) For other services: as per TEMA and Licensor standard.

4.3.2 Fluid Allocation

To allocate fluids to shell or tube side of an exchanger, the following general principles of fluid allocation shall apply:

- 1) cooling water on tube-side
- 2) high pressure fluid on tube-side
- 3) most corrosive fluid on tube-side
- 4) higher fouling fluid on tube-side
- 5) most viscous fluid on tube-side
- 6) large volume of condensing vapours on shell-side
- 7) single phase fluids both sides put smaller flow on shell-side

The above principles may conflict in some instances and alternative designs shall be investigated. In these cases the most economical design shall be selected.

4.3.3 Heat Exchanger Pressure Drops

The pressure drop available shall be specified as below and all hydraulic calculations shall allow for this specified pressure drop.

–Liquids

Total Pressure Drop (bar) for Liquid Flow Through Exchangers (1)			
Viscosity cP @ avg. Temperature	One Shell Total Pressure Drop (bar)	Two Shells in Series Total Pressure Drop: (bar)	Three Shells in Series Total Pressure Drop (bar)
< 1.0	0.35-0.7	0.35 - 0.7	0.7 -1.0
1.0 - 5.0	0.7	1.0	1.0-1.35
5.0 - 10.0	1.0	1.0-1.35	1.35
> 10.0	1.35	1.35-2.0	2.0

Note:(1) Same Total Pressure Drop (bar) for Liquid Flow Through plate fin exchangers.

–Gases

Pressure Drop (bar) for Vapour Flow	
Operating Pressure (bara)	Pressure Drop (bar)
0-1.7	Approx. 0.03 - 0.07
> 1.7	0.14-0.35

–Condensers and Reboilers

- 1) Overhead condensers for towers operating above atmospheric pressure
For partial condensers allow 0.14 to 0.35 bar.
For condensers where total isothermal condensation takes place, the pressure drop is usually low or negligible.
- 2) Surface condensers (vacuum service)
Allow 0.004-0.0067 bar for operating pressures about 0.04 bar (abs).

3) Reboilers

For kettle type reboilers the shell side pressure drop is generally termed 'negligible'.
For thermosyphon type reboilers the exchanger pressure drop must be low and is normally in the region of 0.017 to 0.035 bar.

4.3.4 Airfin Cooler Pressure Drops

The pressure drop available for airfin coolers shall be specified as below and all hydraulic calculations shall allow for this specified pressure drop.

Service	Operating Pressure (bara)	Allowable Pressure Drop ; (bar)	
Liquid Cooling	All	0.7	not valid for viscous fluids
Gas Cooling	2.0-4.5	0.07	
	4.5-18.0	0.21	
	18.0-100	0.35	
Total Condensation	Atmospherics & above	0.03 min	For multi-pass air coolers, high-drops assure proper flow distribution. higher pressure drop will also assure good flow distribution at lower than design throughput
Partial Condensation	Atmospheric & above	0.14-0.35	Note as for total condensation
Condensation	Vacuum	0.004-0.0067	Selection of an allowable pressure drop should be from the results of an economic study

–Air vs. Water Cooling

Air cooling shall be maximised in order to limit the consumption of raw water. As a general rule, the cut-off temperature between air and water cooling shall be 55 °C. If appropriate, the cut-off temperature criteria can be relaxed on a case-by-case basis to avoid the installation of a trim cooler.

The design air inlet temperature shall be 38°C dry bulb. Design dry bulb temperature for air coolers and equipment critical to plant integrity shall be 43°C.

4.4 Pumps

The available NPSH shall exceed the required NPSH by 0.6m at rated flow rate. For Boiler Feed Water Pumps the available NPSH shall exceed the required NPSH by a minimum 2.0m margin up to design capacity at the initial calculation stage. When accurate pump suction layouts are known this margin may be reduced to 1.5m following review of the calculation.

The possibility of pump cavitation should be excluded in all possible operating range of the pumps.

Process engineers shall identify on the process specification turndown flows below 50% of the design capacity, when such flows are possible during long-term operating conditions covered by guarantees of plant performance.

The reference levels for setting NPSHa shall be the bottom tangent line for vertical vessels, the bottom of the vessel for horizontal vessels, the low low level for tanks and the pump impeller centre line for pumps.

The following elevations for pump centreline shall be assumed before pump vendor details are available:

m3/h	Center Line elevation
Up to 45	0.76 m
45 -225	0.91 m
225-2270	1.07 m
2270-4540	1.37 m

Final hydraulic check will be performed with vendor data and final isometrics. A minimum flow bypass will be foreseen for centrifugal pumps for the following cases:

- 1) Differential pressure pumps higher than or equal to 35 bar
- 2) Large pumps with driver power higher than 160 kW
- 3) For process reason (turndown), flowrate lower than or equal to 30 per cent of max flowrate (note : a final comparison between the low process flow and the minimum flow pump protection has to be made with data given by pump vendor regarding the required minimum flow)

To protect the pump, the following systems can be used:

- 1) Flow control valve in the minimum flow recycle line
- 2) Restriction orifice in the minimum flow recycle line (permanent recirculation flow)
- 3) Automatic spill valve with adjacent minimum flow recycle line : this valve opens its side discharge to the pumps suction vessel / system of the discharge flow drops beneath the minimum flow

The choice of the configuration will be done on a case by case study. Nevertheless, except for particular cases, the protection with a permanent recirculation flow is usually adopted for pumps capacity less than 15 m³/h.

The pump data sheet will specify the process flow without provision for the minimum flow which will be specified by the pump's vendor. A note will be added on the process data sheet accordingly.

4.5 Line Sizing

4.5.1 Liquid Flow

The guidelines shown in the following table should be used to size process piping for liquids (for slurries see 4.5.6). The guidelines cover most normal situations for systems within unit battery limits, but they may not be applicable for all cases. For critical services and long headers, the total pressure drop in the system must be checked to ensure the system meets the design pressure balance, whether or not individual process lines meet the pressure-drop and velocity criteria given here.

SIZING LIQUID LINES	'Recommended' High Limits	
	ΔP per 100 m, Velocity (4, 5) barm/s	
Pump suction lines (1) (6)		
- bubble point liquids	0.11	1.8
-subcooled liquids:		
•(< DN 200) [< 8"]	0.45	2.4
•(>DN 200)[>8"]	0.45	3.7
Pump discharge lines		
-CS	0.91	6.1
- alloy / SS	0.91	7.6
Reboilers		
- trapout lines (3)	0.07	1.5
- return lines	0.07	-
Liquid transfer lines (2)	0.34	3.7
Cooling water lines		
- Main header	0.15	3.7
- Branches	0.30	3.7
Steam condensate lines (liquid)	-	0.61

Notes:

- (1) Pump suction line diameters should normally not be more than two (2) standard line sizes larger than the pump suction nozzle.
- (2) Or as required by system pressure balance.
- (3) For sizing of tower draw-offs, refer to section 4.5.2 below
- (4) If the liquid velocity is too high, swaged up meter runs may be required. Velocities for lines containing orifice plates should be limited to the following values:
 - for 2" and up, velocity < 3.4 m/s, except
 - for 14" and up, velocity < 3.1 m/s.
- (5) For reciprocating pump, acceleration head loss has to be taken into account in the pump suction circuit design (the installation of a damper may be required to avoid acceleration loss. To be studied with pump vendor).

4.5.2 Tower Draw-Off Line Sizing

The line size from a drawoff pan is critical for the following reasons:

- 1) Liquid from a tower tray is aerated to some extent depending on the foaminess of the gas-liquid mixture.
- 2) The depth of the drawoff pan is limited due to the interference with the tray below, thus limiting the head of liquid above the drawoff nozzle.

The recommended method for sizing drawoffs employs the following criteria:

- 1) The depth of the drawoff pan to be 1 – 1½ times the nozzle diameter. The minimum allowable depth is 8".
- 2) Allowable velocity may vary from 0.7 m/s to 1.2 m/s depending on the nozzle size (see following tables -'Capacities of Side-Pan Drawoff Nozzles' and 'Capacities of Bottom-Pan-Drawoff Nozzles').
- 3) The nozzle is to be swaged down to a line size which will not exceed 0.1 bar /100m pressure drop. The swage is to occur at a point in elevation 1.2 m below the nozzle drawoff. Only lines 8" and larger are to be swaged down, small lines will be maintained at nozzle size to the pump or first exchanger (see Table 'Typical Swaged Lines After Side-pan Drawoff Nozzle').

Capacities Of Side-Pan Drawoff Nozzles		
Nominal Line Size mm	Allowable Velocity m/s	Flowrate m3/h
80	0.73	12.5
100	0.73	21.6
150	0.73	50.4
200	0.76	88.1
250	0.84	153
300	0.91	237
350	0.98	305
400	1.04	426
450	1.10	567
500	1.14	735
600	1.22	1,027

Capacities Of Bottom-Pan Drawoff Nozzles		
Nominal Line Size mm	Allowable Velocity m/s	Flowrate m3/h
80	0.73	12.5
100	0.73	21.6
150	0.76	52.5
200	0.84	97.4
250	0.91	172
300	0.99	256
350	1.10	344
400	1.16	475
450	1.23	643
500	1.30	838
600	1.37	1,272

Typical Swaged Lines after Side-Pan Drawoff Nozzle :

- 1) Capacities of lines as per 'Capacities of Side-Pan Drawoff Nozzles' table (above);
- 2) Allowable pressure drop limit is approx. 0.1 bar per 100 m.
- 3) Assumed drawoff fluid properties: hot SG = 0.8; viscosity ≤ 3 cSt.

Nominal line size mm	Assumed Flowrate m³/h	Swaged line size mm	Pressure drop in swaged line bar per 100 m	Velocity in swaged line m/s
80	12.5	80	0.07	0.7
100	21.6	100	0.05	0.7
150	50.4	150	0.04	0.7
200	88.1	150	0.10	1.3
250	153	200	0.07	1.3
300	237	200	0.12	2.0
350	305	250	0.08	1.7
400	426	300	0.07	1.6
450	567	300	0.10	2.2
500	735	350	0.11	2.3
600	1,027	400	0.11	2.8

Notes: Swaged line size may be slightly different depending on physical properties of fluid, static head, physical layout and position of swage in relation to drawoff nozzle.

4.5.3 Vapour Flow

The guidelines shown in the following table should be used to size vapour lines. The guidelines cover most normal situations for systems within unit battery limits, but they may not be applicable for all cases. For critical services and long headers, the total pressure drop in the system must be checked to ensure the system meets the design pressure balance, whether or not individual process lines meet the pressure drop and velocity criteria given here. For long vapour lines, such as flare headers or vacuum transfer lines, when the $\Delta P > 10\% P$, the compressible flow calculation procedure should be adopted.

SIZING VAPOUR LINES		'Recommended' High Limits	
Operating Pressure Range		Maximum	Maximum
		□ P per 100 m bar	Velocity (1) m/s
HYDROCARBON LINES			
Vacuum: 0.07 bara or less		0.014	$122/\rho^{0.5}$
Vacuum: ~ 0.48 bara		0.034	$122/\rho^{0.5}$
0.0-3.45 barg		0.11	$122/\rho^{0.5}$
3.45-10.3 barg		0.34	$122/\rho^{0.5}$
10.3-34.5 barg-		0.68	$122/\rho^{0.5}$
> 34.5 barg		1.13	$122/\rho^{0.5}$
STEAM LINES (< 90 m in length)			
Vacuum: 0.07 bara or less		0.014	-
Vacuum: - 0.48 bara		0.034	-
0.0-3.45 barg		0.11	-
3.45-10.3 barg		0.34	-
10.3-34.5 barg-		0.68	-
> 34.5 barg		1.13	-

Notes:

(1) Vapour density (ρ): measured in kg/m^3

4.5.4 Slurry lines

Slurry lines are to be designed on a case by case basis. Solid suspension and erosion shall be considered in the design.

4.6 Control Valves

Control valves shall be sized for a normal flow at no more than 70% of the capacity of the valve.

The control valve pressure drop should at least be equal to the maximum value calculated by the following procedures at the stated flowrate :

At normal flowrate,

Control valve pressure drop > 25% of Total Frictional Pressure Drop or...

Control valve pressure drop > 33% of Frictional Pressure Drop, excluding CV DP.

At normal flowrate,

Control valve pressure drop > 10% of Discharge Vessel Pressure, for vessel operating pressure

< 70 bara

Above 70 bara a lower percentage of the operating pressure may be used for valve differential pressure depending on process and control considerations for non flashing services.

At design flowrate,

Control valve pressure drop > 0.7 bar for general case

Control valve pressure drop > 0.2 bar for butterfly valve

Exceptional cases (e.g. gravity flow) where lower pressure drops are required should be evaluated on a case by case basis.

Control valves in continuous service, of a nominal size 6" or less, shall generally be provided with isolation and bypass valves to allow maintenance or repair without shutting down the plant.

Bleeds shall be provided on the upstream side (or both upstream and downstream side) of the valve as appropriate. Control valves of size 8" or larger shall be fitted with hand wheel rather than a bypass, if economically justified. On a case-by-case basis, it may be decided not to install by-pass for operation or safety considerations (e.g. letdown from very high pressure, etc.). In such a case, a note on the PID should specify the absence of bypass.

4.7 Equipment Design Life

4.7.1 Design Life

The required minimum design life for equipment is defined in the Basic Engineering Design Data doc.

4.8 Equipment Design Temperature

4.8.1 Upper Design Temperature

Design temperature shall be set at 25°C above the maximum operating temperature. The minimum upper design temperature shall be 80°C.

The maximum operating temperature is defined as the highest material balance temperature. Consideration shall also be given to other conditions such as startup, shut-down, catalyst regeneration, failure of upstream cooling and steam-out. Such temperatures should be considered on a case by case basis to establish if they should be taken as the maximum operating temperature.

In the case of coolant failure for process units, the maximum operating temperature upstream of the cooler shall be taken as the downstream design temperature. This

should apply to downstream pipework, only up to the next major item of equipment. For failure of rundown coolers, the higher design temperature shall only be taken to the battery limit. A high temperature alarm shall be installed downstream of all product rundown coolers.

4.8.2 Lower Design Temperature

The minimum mechanical design temperature for which process equipment must be designed will normally be specified as the lower of:

- 1) Minimum operating temperature of the equipment minus 5°C
- 2) Minimum mechanical temperature resulting from ambient temperature.

In generally, the minimum design temperature of pressurized equipment operated above ambient temperature shall be -28°C.

The minimum design temperature of pressurized equipment operated at ambient temperature shall be -38°C of absolute ambient temperature.

- 3) Minimum abnormal operating temperature that could occur during an abnormal mode of operation such as start-up, shut-down or emergency de-pressuring, but only need be quoted for values less than 0°C. No margin shall be taken on this abnormal low temperature. The associated pressure shall be indicated (put a note with the couple temperature / pressure) or, if required, the curve pressure - temperature shall be attached.

For depressurisation of fluids which have an atmospheric boiling temperature below 0°C the following should apply:

1) Liquid Filled Systems

- a) Minimum temperature shall be taken as the resulting boiling point after adiabatic expansion of the liquid from maximum operating pressure to atmospheric pressure.
- b) For multicomponent mixtures the minimum temperature shall be determined by flashing the fluid from operating to atmospheric pressure in a series of flashes, removing the vapour after each stage of flashing.

2) Gas Filled Systems

Minimum temperature is to be taken as that resulting from a 40% efficient isentropic expansion from maximum operating pressure to atmospheric pressure.

If an equipment item can be depressurized, and there of it immediately re-pressurized, its mechanical design conditions shall include its mechanical design pressure calculated with the auto-refrigeration temperature.

4.8.3 Multiple Design Temperature

When different metal temperatures can be predicted to occur for different zones of a vessel during operation (as in the case of a large distillation column) then the different temperatures zones should be indicated on the vessel sketch in the vessel

specification and these different temperatures should be taken into account for setting design temperatures for each zone.

4.9 Equipment Design Pressure

When several pieces of equipment are protected by the same relief valve, each piece of equipment will be designed, at least, for the pressure imposed by the discharge conditions of the relief valve in case of emergency.

For example, column reflux drums are to have a design pressure 0.5 bar (to be checked and adjusted with final plot plan) higher than the column design pressure. This is to account for back-up of liquid to the condenser in case of loss of reflux.

4.9.1 Pumped Systems

Process piping and equipment which form part of a pumped hydraulic system, and may operate liquid full, will normally be designed for the maximum pressure that can be developed by the pump.

For centrifugal pumps for a single consistent case.

Maximum Discharge Pressure	=	Maximum Differential Pressure	+	Maximum Suction Pressure
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Where:

Maximum Suction Pressure	=	Suction vessel Design Pressure (or, Relief Valve set pressure)	+	Vessel Static Head @ HLL
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Pressure drop across trays or vessel internals should be included if it is significant.

Maximum Differential Pressure	=	F	X	Net differential Pressure (@ Rated Capacity)
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Where:

F is (before vendor data is available): F

1.25 for motor driven pumps

1.38 for turbine driven or variable speed pumps

Note: Calculation is based on maximum operating S.G.

A final check will be performed with vendor data.

Reciprocating pumps shall be provided with discharge relief valves, the pressure of which shall be set to avoid the pump overpressuring any system into which it discharges.

4.9.2 Compressor Systems

For centrifugal compressor systems, consideration should be given to the design of the suction system for settle-out pressure at shut-down. Under these circumstances high pressure gas from the discharge side will be passed to the suction side through the anti-surge recycle valve.

The settle out pressure will be a function of the relative volumes of the suction and discharge systems.

Reciprocating compressors shall be fitted with a discharge relief valve set to prevent the compressor overpressuring any system into which it discharges.

4.9.3 Non-Pumped Systems

The design pressure for equipment in non-pumped systems shall be set as follows:

Max. Operating Pressure (MOP) barg	Design Pressure barg
≤ 0.3	Set on a case by case basis
$0.3 < \text{MOP} < 17$	$1.7 + \text{MOP}$
$17 \leq \text{MOP} < 40$	110% of MOP
$40 \leq \text{MOP} < 80$	$4.0 + \text{MOP}$
≥ 80	105% of MOP

For all process equipment (with the exception of storage tanks and certain offsites facilities) the minimum design pressure shall be 3.5 barg.

Derogations allowing lower mechanical design pressures can be given for specific licensor designed equipment items.

4.9.4 Shell and Tube Heat Exchangers

For shell and tube heat exchangers where the design pressure of one side is considerably higher than the other, the design pressure of the lower pressure side shall be 77% of that of the higher side. This will mitigate the need for provision of relief facilities to protect the heat exchanger on the lower pressure side in the event of a tube rupture.

The impact of the tube rupture on the upstream and downstream low pressure circuit shall be evaluated.

4.9.5 Steam-out conditions

Steam-out conditions are to be 1.5 barg @ 140 °C. If steam out connections are permanent, steam-out conditions are to be Full Vacuum @ 140°C.

All process specifications for equipment that shall be steamed-out, shall clearly state this requirement and the steam-out conditions.

4.10 Vessels subject to Vacuum Conditions

Vessels which may operate under vacuum conditions during normal operation shall be designed for full vacuum.

Design Pressure and Temperature Example

With reference to Fig. 1 (below):

Vessel design pressure:

P_{di} = Relief valve set pressure, P_{r1} , P_{di} , P_{r1} , However, for liquid filled systems,

$P_{r1} = P_{d1} - P_{h3}$

Pump suction:

$P_{02} = P_{01} + P_{h1} - \text{Frictional Losses (based on normal SG)}$ $P_{02} = P_{01} + P_{h1} -$

$P_{d2} = P_{d1} + P_{h1} + P_{h2}$ (based on maximum SG) $P_{d2} = P_{d1} + P_{h1} + P_{h2}$

For pumps in parallel, ie. one operating, one standby, the downstream design pressure is taken back to, and including, the suction block valve.

Pump discharge:

P_{03} is Net Discharge Pressure P_{03}

P_{d3} is Maximum Discharge Pressure P_{d3}

$P_{d3} = \text{Maximum Pump Differential Pressure} + P_{d2}$ $P_{d3} = + P_{d2}$

P_{d3} should be checked against the design pressure of the downstream equipment. If, for example, a downstream column has a design pressure higher than expected, say to contain a runaway reaction, this pressure may be taken as the design pressure back to a suitable break point, say the pump's suction block valve.

The method for setting the control valve pressure drop in a pump's discharge circuit is given in Section 5.6 (Control Valves).

In certain circumstances the low pressure side design pressure may be increased to 77% of the high pressure side design pressure to allow for the burst tube case as defined in API Standard 520 and API Standard 521. Certain codes dictate the provision of relief protection on heat exchangers.

Top: Pd6 = Pr2

Pd7 does not include the liquid static as this will be added by the vessel design group based on the high level.

$$\text{Pd8} = \text{Pd7} + \text{Ph4} + \text{Ph5}$$

FIGURE 1 EXAMPLE PRESSURE PROFILE FOR A PUMPED SYSTEM

The diagram illustrates a pumped system with the following components and labels:

- Storage Tank (Left):** Contains liquid levels HLL, NLL, and LLL. The total head from the reference point to the tank is $H_1 (P_{h1})$. The pressure at the top is P_{o1} and P_{d1} . A break point for pumps in parallel is indicated.
- Pumps:** Two pumps are shown in parallel, with a break point for a single pump indicated. The pressure at the pump outlet is P_{o2} and P_{d2} .
- Series Pump:** A single pump in series with the parallel pumps, with pressure P_{o4} and P_{d4} .
- Vertical Column (Right):** The liquid level is HLL. The pressure at the top is P_{o12} and P_{d12} . The pressure at the bottom is P_{o10} and P_{d10} .
- Head Differences:** $H_2 (P_{h2})$ is the head from the tank level to the pump inlet. $H_3 (P_{h3})$ is the head from the tank level to the pump outlet. $H_4 (P_{h4})$ is the head from the pump outlet to the column level. $H_5 (P_{h5})$ is the head from the pump outlet to the column bottom.
- Legend:**
 - P_o = OPERATING PRESSURE
 - P_d = DESIGN PRESSURE
 - P_h = PRESSURE DUE TO STATIC

DPTDFIG1.DRW

5.1 Vent of equipment and pipe

Vent shall be provided for columns, vessels and pipe with special requirement, to be used for disposing failure、purging of shutdown and steam out of startup.

The following list is a guide for sizing minimum vent pipe size of pipeline:

Pipe size(d) mm	Vent pipe size mm
<20	15
25<d<150	20
200<d<350	25
>400	40

Vent to atmosphere of equipment and pipe shall be satisfied with environment and safety requirement, if not , vent shall be collected to recycle or discharged to flare.

Vent shall be provided for equipment with manhole, at the top of equipment or nearby, considering breathing of operator inside. Vent and manhole shall be set at both ends of horizontal vessel.

5.2 Drain of equipment and pipe

The drain (such as oil, solvent(s) or chemical reagent(s)) due to sampling, overflow, accident or from low point of piping shall go to closed collection system or other collection facility. It's not allowed to drain locally or into sewer system.

Plant shall be provided collection drum according to operation requirement to collected liquid drain from such as sample point and low point , then routing to corresponding system by free draining, discontinuously inert gases pushing or pumping. When plant shutdown as a result of failure or operating reason, drain shall be routed to corresponding tankfarm through normal operational pipe.

All equipment in plant shall be provided low point drain, normally at bottom of vessels, the pipe connected with column、exchange or heater have block valve, shall be provided drain before the valve, if it is not, shall be provided at lowest point。The size of drain pipe shall be according to following list, equipment like air cooler or pump shall be provided drain screw plug, but it is not applied for fluid of high temperature.

Drain shall be provided for equipment and pipe with fluid concreted and freezing easily

Drain pipe shall be provided between contral valve and upstream block valve.

Liquid drain with high temperature (electric desalting water, drain of waste heat boiler, etc) shall be cooled before routed to sewage system. The maxiumm temperature drain to sewage is 60°C.

Large size pipe(crude oil pipe.etc) shall be provided low point drain for purging, the drain pipe shall be fitted with double vavle or a vavle and a spectacle blind behind

5.3 Vent and drain valve

Generally gate valve is used for vent and drain pipe. The ball valve, plug vavle and butterfly valve with soft seal are unallowed for vent and drain of flammable and toxic fluid to prevent greater leak due to melt of soft seal in case of external fire.

System saturation vapor pressure above 5 barg at 40°C, vent and drain pipe for system shall be fitted with double valve.

The alloy steel material shall be used between alloy steel equipment and first block valve (inclusive), CS could be used after the first block valve.

The following list is requirement of vent valves, drain valves and pipe fittings:

System	Operation requirement		
	Valve open for normal or frequently operation	Used for start-up, shutdown or equipment isolation	Only used for pressure test
Hazard material	1. Only provide vent and drain valve. 2. Vent and drain pipe shall be satisfy with the requirement.	Vent and drain valve with plug, cap and blind.	1. Vent pipe shall equipped with plug, cap and blind. 2. Drainpipe shall equipped with plug, cap and blind.
Steam operation pressure less than 20 bar	Only provide vent and drain valve.		
Non-hazard material	Vent and drain valve with plug, cap and blind.		

Notes :

- 1)Toxic and high corrosion material, should be discharge closed system based on material property.
- 2) Size of plug and cap 1 1/2"(DN40); Size of blind 2"(DN50)
- 3)Flammable liquid should be discharge closed system or safety location based on requirement.
- 4)If steam condensate recycle system is not available, condensate from steam trap shall not be discharged to pavement or other area operator usually pass by.
- 5)Non-hazard material discharge, could use drain valve.

6 DISPOSAL SYSTEMS

Fluids discharged from pressure-relieving devices shall be disposed of to one of the following locations:

- another part of the process at lower pressure
- atmosphere
- flare
- blowdown header
- oily water sewer.

Burn pit (for alkyl systems - if applicable) The discharge location shall be selected on the basis of the phase, composition and hazardous nature of the fluid.

In all cases, the backpressure at the relieving devices should be applied.

6.1 Discharge to the Process

This is the preferred disposal route, since the discharged fluid is retained within the process unit. It shall be considered in all instances where the flow is from a pump discharge (relief is typically routed to pump suction). It is particularly appropriate for liquid discharges, corrosive fluids and fluids which are otherwise difficult to dispose of. However, opportunities which do not create the potential for overpressure elsewhere in the unit are expected to be limited.

6.2 Discharge to Atmosphere

To safeguard the environment, normal atmospheric discharges shall be limited to non-flammable and non-toxic fluids - typically steam, air; nitrogen or water. Any other essential requirement to discharge hydrocarbons or any other flammable or toxic stream to atmosphere shall be highlighted at the earliest opportunity and agreed with Company.

The discharge point for an atmospheric release shall be at a safe location with the flow directed away from areas where personnel may be present.

To ensure adequate dispersion, the discharge line shall be sized to provide a minimum exit velocity of 152 m/s (500 f/s) with a maximum limit of 0.80 Mach. The minimum line size shall be the diameter of the relief valve discharge nozzle. If appropriate, the size of the downstream section of the line shall be reduced to enable the minimum exit velocity to be achieved. If the fluid is flammable or toxic, a dispersion analysis shall be conducted to confirm safe disposal. The discharge line shall be otherwise sized to limit the backpressure at the relief valve.

The calculated pressure drop shall include the tail pipe exit loss and be based on the rated flow through the valve and a pipe roughness of 0.046 mm for carbon steel pipe. This value reflects the potentially corroded nature of a relief valve tail pipe.

No isolating valve shall be installed in the tail pipe. The tail pipe shall be provided with a low point drain to allow any accumulated liquid to be removed, together with a suitable cap to prevent the ingress of rain or foreign bodies.

A snuffing steam connection shall be provided if there is the potential for the discharge to be ignited. Flame arrestors shall not be utilized to prevent flashback.

During the detailed engineering phase, assessments shall be made of: the noise generated when the relief valve is discharging at the maximum anticipated rate. If appropriate, a silencer shall be provided to reduce noise to an acceptable level. Account shall be taken of any additional pressure drop introduced by the silencer the impact of thermal radiation in the event that a flammable vent is ignited.

Storage tanks containing flammable or toxic fluids shall be protected against overpressure using pressure-vacuum valves. The out-breathing of volatile materials shall preferably be routed to flare and shall only by exception be routed to atmosphere. A dispersion analysis shall be performed to confirm that any flammable or toxic vapour discharged to atmosphere will be disposed of without risk to personnel or equipment.

6.3 Discharge to Flare

The flare system is primarily intended to collect and destroy emergency relief streams occurring in case of incidents or process upsets. All relief devices capable of relieving flammable or toxic fluids shall, wherever feasible, discharge to a flare system.

No flaring of operational relief streams is allowed in normal operation. Operational flaring shall be avoided as much as possible, and limited to cases where alternative handlings are not practicable.

6.4 Discharge to Blowdown Header

Thermal relief valves discharging hydrocarbons or flammable or toxic material shall discharge, wherever possible, to a dedicated blowdown header, from where they shall be routed via a blowdown drum to flare (see section 6). Manual blowdown valves used to depressure and drain equipment prior to maintenance shall also discharge to this location.

6.5 Discharge to Oily Water Sewer

Thermal relief valve discharges which are non-flammable, non-flashing and non-toxic shall discharge, wherever possible, to a lower pressure process system. If no suitable location is available, the discharge shall be routed to the oily water sewer or grade, as appropriate.

6.6 Discharge to a Burn Pit

Pressure relief vents from the alkyl systems shall be vented to the alkyl scrubber and not be vented to the flare. Where the alkyl is diluted to a point where it is considered as non pyrophoric it may be vented to the flare system designed for alkyl / flammable mixtures.

The vent from the alkyl scrubber shall discharge to a burn pit. The burn pit shall consist of approximately 0.5 m³ of large gravel or equivalent and be located in a safe area outside the curbed area. The burn pit shall be kept sheltered against rainfall (Chinese hat) and designed to drain away any possible water. A barrier shall be installed around the burn pit to prevent access. The vent pipe shall terminate 0.1 m above the gravel in the burn pit.

By-passing flow around equipment or piping which is temporarily or seasonally out of service

7 SPECIFICATION FOR INSULATION AND TRACING

Design of insulation and tracing for equipment and pipe generally refers to design of hot insulation, cold insulation, people protection and winterization.

7.1 Hot insulation

Hot insulation shall be provided for equipment and pipe on following situation:

- Hot insulation for process requirement;
- Equipment and piping with fluid of freezing point above ambient temperature;
- Condensate due to ambient temperature will corrupt equipment and piping;
- The thermal insulation (to prevent burns) pipes shall be provided at the pipe wall outside the working area or maintained area above 60°C, and in the workplace and in the service area at a temperature above 45°C.

Equipment, pipe and accessories without hot insulation in operation and the appearance temperature exceeds 60°C , whereas needed for regular operation which can not use other measures to prevent scalding, shall be provide person protection to ensure operator safety, and improve working conditions.

The height less than 2.1m from the ground or platform

The distance less than 0.75m close to the platform

Refrigeration system cooling equipment, cooling pipes and accessories, needed to reduce the cooling loss and prevent cooling medium higher temperature The equipment or piping with fluid temperature below ambient dew point temperature, needed to prevent dew of outer surface The equipment or piping with potential

evaporation caused by sunshine or the outside temperature, such as LPG pipeline at suction section of pump, often exposure in the sunshine.

7.1.1 Location without hot insulation

Hot insulation is not necessary for following situation:

Equipment and pipe of rejection of heating:

Equipment and pipe need to detect leak in time

Equipment and pipe with internal hot insulation and wearable lining

The location need to inspect frequently to avoid damaging:

Vent and drain of equipment and pipe is not need hot insulation in operation

8 UTILITY STATION

Utility station used in supplying materials for temporary cleaning, replacement, start-up preparation, incident handling and maintenance, shall not be used for normal operation.

The following shall be considered for utility station:

Utility station design shall be based on different process conditions and requirements and material properties.

Plant air shall be used for utility station, and instrument air is unallowed.

Fire water, steam with pressure higher than 10 bar , cooling tower water and steam condensate shall not be used for utility station. Location of utility station should be considered to ensure pipeline can still be used even if a section or unit shutdown.

9 EQUIPMENT SPARING

9.1.1 General

Installation of standby spare equipment shall be included where justified for safety, reliability or economic reasons. All units are scheduled for turnaround once every five years for planned maintenance. Running equipment requiring more frequent turnarounds shall justify a spare.

Whilst aiming to minimise standby equipment care should however be taken to ensure that unit operations are not rendered vulnerable to failure of relatively

inexpensive equipment (e.g. lube oil and seal oil pumps for large turbo-compressor units).

9.1.2 Heat Exchangers

Heat exchangers with a high fouling potential shall be arranged in parallel trains, such that one train can be taken out of service for cleaning while the train in operation meets the plant's minimum capacity.

9.1.3 Pumps

All pumps with an immediate influence on the process must have a spare. However, consideration shall be given to the use of common installed stand-by pumps performing two duties.

Sparing of pumps with a delayed influence on the process (e.g. inhibitor feeding pumps) will be evaluated on an individual basis.

In general, both the main pump and the spare are to be motor driven. However, consideration is to be given to steam turbine drive for main and/or spare pumps where significant emergency relief load reduction can be achieved.

Consideration shall be given to having steam turbine drives and motor drives in critical services, such as BFW. For critical services such as firewater, the use of diesel drives on pumps shall be considered. Refer to specific Basis Of Design documents.

9.1.4 Compressors, Fans, Blowers and Filters

Sparing of compressors will be as follows:

- Centrifugal compressors shall have a spare rotor in the warehouse.
- Reciprocating compressors shall have a 100% installed spare,

For fired heaters fans:

ID fans shall not be spared.

FD fans shall have a 100% spare.

Filters:

A spare filter shall be provided for filters in continuous service.

10 VALVE LEAKAGE CLASSIFICATION

All control valves and block & bleed valves connected to a flare system shall be of leakage 'Class V' or better.

11 OVERDESIGN MARGINS

As a general rule, the following oversize margins shall be taken for the sizing of individual equipment items. For pumps and compressors in licensed process units, oversize margins proposed by the licensor will be retained where larger than specified hereafter. For other materials, the design specified by licensor will be applied as is and supersede the oversize margins hereafter.

Pumps:

The maximum operating flow rate is taken as base for the pump design. The following margins are applied to determine the rated flow of the pump:

- + 10 % in general process or utility services
- + 10 % in reboiler feed services
- + 20 % in column reflux services

Compressors:

The maximum operating flow rate is taken as base for the compressor design. The following margins are applied to determine the rated capacity:

- + 10 % in general process or utility services

Tubular heat exchangers:

Thermal design based on 110 % of maximum operating flow rates (with same temperature profile) and corresponding heat exchange duty.

Hydraulic design based on 110 % of maximum operating flow rates, when the exchanger is at a pump or compressor discharge.

Overhead condensers of fractionation columns shall be designed (thermal and hydraulic) for 120 % of regular operating flow rates and heat exchange duty.

Plate heat exchangers:

Thermal design based on 105 % of maximum operating flow rates (with same temperature profile) and corresponding heat exchange duty.

Hydraulic design based on 105 % of maximum operating flow rates, when the exchanger is at a pump or compressor discharge.

Air coolers:

Same oversize margins as for tubular heat exchangers.

Heaters:

As defined by the licensor of process units. No process heaters are installed in the utility or off-site facilities of the Power Plant.

Drums:

In sizing for liquid-vapour separation, liquid-liquid settling of solid particle decantation, no additional margin is taken over the sizing calculation result.

Hold-up volumes in a drum are based on the hold-up time requirements specified in section 4.1, without additional margin.

The resulting drum dimensions shall be rounded up to the next higher 50 mm, if less than or equal to 2000 mm, and to the next higher 100 mm if larger than 2000 mm.

Column internals:

Design based on the governing case of the material balance and on the specifications of section 4.2 of this document. No additional oversize margin.